

```
module CalculatorApp {
```

```
    exception DivisionByZero {};
```

```
    interface Calculator {
```

```
        float add(in float a, in float b);
```

```
        float subtract(in float a, in float b);
```

```
        float multiply(in float a, in float b);
```

```
        float divide(in float a, in float b) raises(DivisionByZero);
```

```
    };
```

```
};
```

```
-----  
import CalculatorApp.*;
```

```
import org.omg.CORBA.*;
```

```
import org.omg.CosNaming.*;
```

```
public class CalculatorClient {
```

```
    public static void main(String args[]) {
```

```
        try {
```

```
            // Initialize ORB
```

```
            ORB orb = ORB.init(args, null);
```

```
            // Get reference to Naming Service
```

```
            org.omg.CORBA.Object objRef =
```

```
orb.resolve_initial_references("NameService");
```

```
            NamingContextExt ncRef = NamingContextExtHelper.narrow(objRef);
```

```
            // Resolve Calculator object
```

```
            Calculator calculator =
```

```
CalculatorHelper.narrow(ncRef.resolve_str("Calculator"));
```

```
            // Perform operations
```

```
            System.out.println("Addition (10 + 5): " + calculator.add(10, 5));
```

```
            System.out.println("Subtraction (10 - 5): " + calculator.subtract(10,  
5));
```

```
            System.out.println("Multiplication (10 * 5): " +  
calculator.multiply(10, 5));
```

```
            try {
```

```
                System.out.println("Division (10 / 2): " + calculator.divide(10,  
2));
```

```
                System.out.println("Division (10 / 0): " + calculator.divide(10,  
0)); // Should raise exception
```

```
            } catch (DivisionByZero e) {
```

```
                System.out.println("Error: Division by zero is not allowed!");
```

```
            }
```

```

        } catch (Exception e) {
            System.err.println("Client exception: " + e.getMessage());
            e.printStackTrace();
        }
    }
}

```

```

-----
import CalculatorApp.*;
import org.omg.CORBA.*;
import org.omg.CosNaming.*;
import org.omg.PortableServer.*;

```

```

class CalculatorImpl extends CalculatorPOA {
    private ORB orb;

    public void setORB(ORB orb_val) {
        orb = orb_val;
    }

    public float add(float a, float b) {
        return a + b;
    }

    public float subtract(float a, float b) {
        return a - b;
    }

    public float multiply(float a, float b) {
        return a * b;
    }

    public float divide(float a, float b) throws DivisionByZero {
        if (b == 0) throw new DivisionByZero();
        return a / b;
    }
}

```

```

public class CalculatorServer {
    public static void main(String args[]) {
        try {
            // Initialize ORB
            ORB orb = ORB.init(args, null);

            // Get reference to RootPOA and activate POA Manager
            POA rootpoa =
POAHelper.narrow(orb.resolve_initial_references("RootPOA"));
            rootpoa.the_POAManager().activate();

```

```

        // Create servant and register it with ORB
        CalculatorImpl calculator = new CalculatorImpl();
        calculator.setORB(orb);
        org.omg.CORBA.Object ref = rootpoa.servant_to_reference(calculator);
        Calculator href = CalculatorHelper.narrow(ref);

        // Bind object to Naming Service
        org.omg.CORBA.Object objRef =
orb.resolve_initial_references("NameService");
        NamingContextExt ncRef = NamingContextExtHelper.narrow(objRef);
        NameComponent path[] = ncRef.to_name("Calculator");
        ncRef.rebind(path, href);

        System.out.println("Calculator Server is running...");
        orb.run();
    } catch (Exception e) {
        System.err.println("Server exception: " + e.getMessage());
        e.printStackTrace();
    }
}

```

```

-----
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```